# ERP Part Intake Tool

*AI-Assisted ERP Automation System*

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# Background

- ERP stands for Enterprise Resource Planning
- ERP systems are used by companies to manage inventory, purchasing, and manufacturing records
- This project was developed around workflows used at igus RBTX during my internship

# Problem Statement

Whenever a new part is purchased from a vendor, it must be set up manually in the system.

- Part creation is often time consuming
- Requires ERP trained associate. Most engineers don't have access to LN.
- Must check for duplicate records
- Manual workflows take unnecessary time

# Objectives/Workflow

Develop a functional Streamlit MVP to:

- Create searchable Parts list
- Generate automatic internal part numbers
- Prevent duplicate manufacturer part numbers
- Web search for parts

# Application Overview

- Built using Streamlit and Python
- Automates internal part setup
- Stores and updates ERP part data
- Standardizes descriptions and numbering

app.py | requirements.txt | README.md

app.py > ...

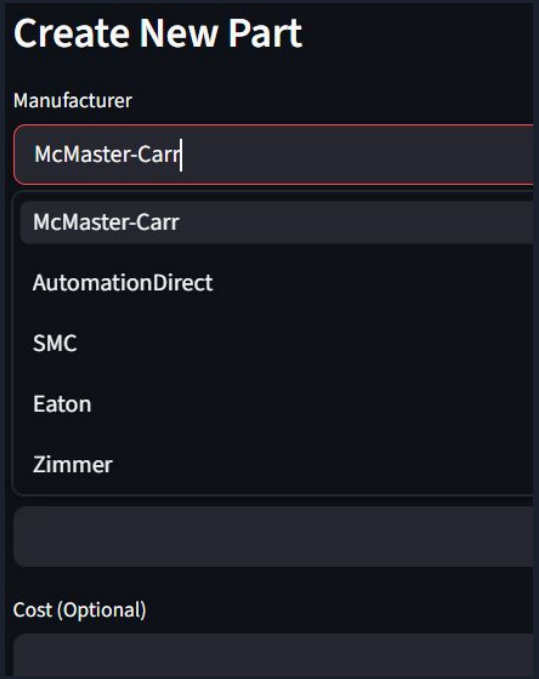
```python
import streamlit as st
import pandas as pd
import re
from datetime import datetime
import os
from urllib.parse import quote_plus

FILE_NAME = "RBTXPartsList.xlsx"


def initialize_file():
    if not os.path.exists(FILE_NAME):
        df = pd.DataFrame(columns=[
            "RBTX PN",
            "Manufacturer",
            "Manufacturer PN",
            "Description",
            "Cost",
            "Date Added"
        ])
        df.to_excel(FILE_NAME, index=False)


def load_data():
    return pd.read_excel(FILE_NAME)
```

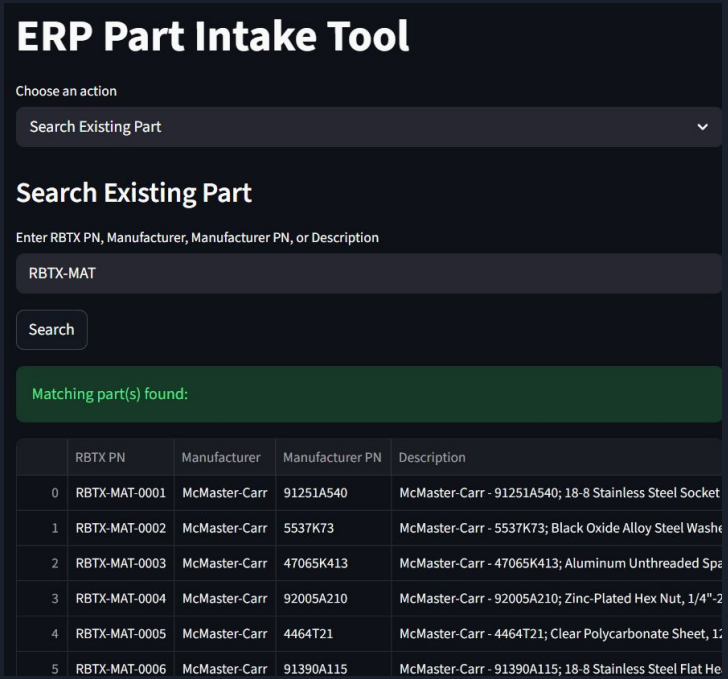
PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | FOLDERS

```
PS C:\Users\cvanh\OneDrive\Documents\EGR404_ERP_Project> py -3.11 -m streamlit run app.py
2026-05-05 16:48:31.393 Uvicorn server started on 0.0.0.0:8501

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.1.161:8501
```

# Key Features

- Automatic internal part number generation
- Vendor-based part categorization
- Duplicate manufacturer part detection
- Standardized ERP descriptions
- Searchable parts database
- Automated vendor web lookup

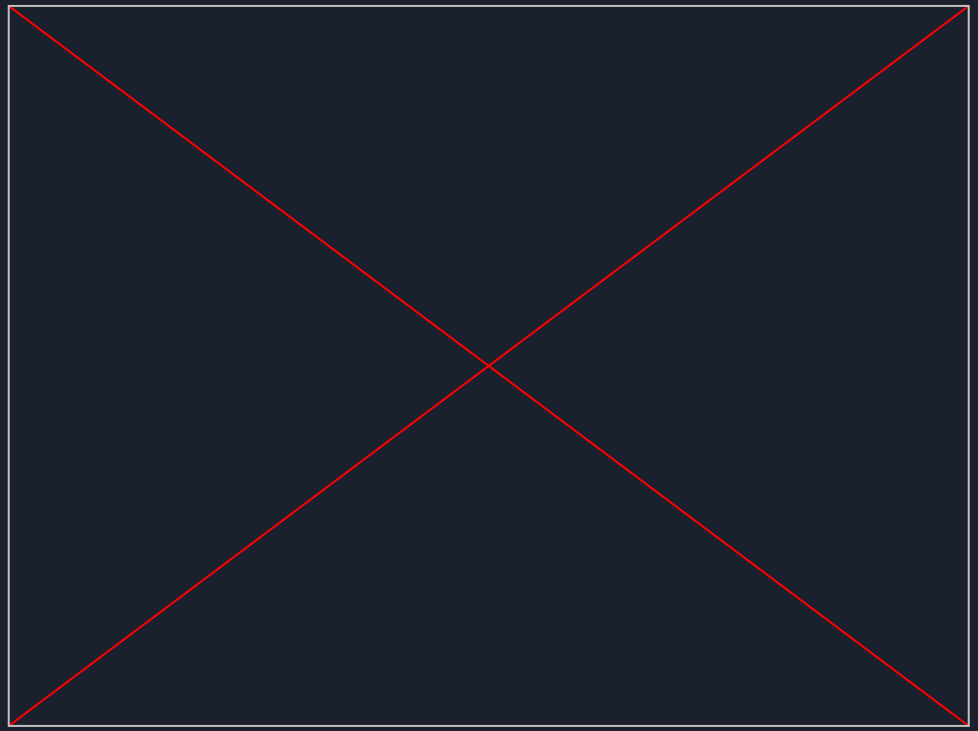


# RBTX Parts List

| RBTX PN | Manufacturer | Manufacturer PN | Description | Cost | Date Added |
|---|---|---|---|---|---|
| RBTX-MAT-0001 | McMaster-Carr | 91251A540 | McMaster-Carr - 91251A540; 18-8 Stainless Steel Socket Head Screw, 1/4"-20 Thread, 1-1/2" Length | 4.82 | |
| RBTX-MAT-0002 | McMaster-Carr | 5537K73 | McMaster-Carr - 5537K73; Black Oxide Alloy Steel Washer, 1/4" Screw Size | 2.15 | |
| RBTX-MAT-0003 | McMaster-Carr | 47065K413 | McMaster-Carr - 47065K413; Aluminum Unthreaded Spacer, 1/4" ID, 1/2" OD, 1" Length | 6.47 | |
| RBTX-MAT-0004 | McMaster-Carr | 92005A210 | McMaster-Carr - 92005A210; Zinc-Plated Hex Nut, 1/4"-20 Thread | 1.08 | |
| RBTX-MAT-0005 | McMaster-Carr | 4464T21 | McMaster-Carr - 4464T21; Clear Polycarbonate Sheet, 12" x 12" x 1/4" | 18.95 | |
| RBTX-MAT-0006 | McMaster-Carr | 91390A115 | McMaster-Carr - 91390A115; 18-8 Stainless Steel Flat Head Screw, #10-32 Thread, 1" Length | 3.64 | |
| RBTX-MAT-0007 | McMaster-Carr | 8975K17 | McMaster-Carr - 8975K17; Steel Shaft Collar, 1/2" Bore Diameter | 7.29 | |
| RBTX-SMC-0001 | SMC | SY3120-5LZD-C6 | SMC - SY3120-5LZD-C6; 5-Port Solenoid Valve, 24VDC, C6 Port Size | 89.5 | |
| RBTX-SMC-0002 | SMC | CDQ2B32-25DZ | SMC - CDQ2B32-25DZ; Compact Pneumatic Cylinder, 32mm Bore, 25mm Stroke | 54.75 | |
| RBTX-SMC-0003 | SMC | AW30-N03BG-Z-A | SMC - AW30-N03BG-Z-A; Modular Air Regulator, 1/4" NPT Port | 42.1 | |
| RBTX-SMC-0004 | SMC | MXQ12-30 | SMC - MXQ12-30; Air Slide Table, 12mm Bore, 30mm Stroke | 118.3 | |
| RBTX-SMC-0005 | SMC | ISE30A-01-N-MGK | SMC - ISE30A-01-N-MGK; Digital Pressure Switch, NPN Output | 67.99 | |
| RBTX-ETN-0001 | Eaton | D-M9BW | Eaton - D-M9BW; Auto Switch Sensor for Pneumatic Cylinder | 14.25 | |
| RBTX-ETN-0002 | Eaton | E50KH3 | Eaton - E50KH3; Miniature IEC Control Relay | 22.8 | |
| RBTX-ETN-0003 | Eaton | M22-PV | Eaton - M22-PV; 22mm Push Button Operator, Green | 11.45 | |
| RBTX-MAT-0008 | McMaster-Carr | 90044A125 | McMaster-Carr - 90044A125 | | 2026-05-03 15:44:34 |
| RBTX-SMC-0006 | SMC | VFN200N | SMC - VFN200N; NAMUR Interface 3-Port Solenoid Valve | | 2026-05-05 15:56:13 |

# Part Search/Creation Demo



# Benefits & Impact

| BEFORE | AFTER |
| --- | --- |
| Manual entry | Automated workflow |
| Inconsistent naming | Standardized descriptions |
| Duplicate parts | Database validation |
| Time-consuming setup | Faster processing |

# Future Improvements

- Direct integration with LN ERP
- Automatic manufacturer part scraping
- Auto-filled descriptions and pricing
- Reduced manual engineering input

# Conclusion

**ERP Part Setup Automation MVP**

- Reduced repetitive ERP setup tasks
- Improved part organization and consistency
- Demonstrated real-world workflow automation
- Built around engineering operations at igus RBTX, and allows our engineers to be involved in ERP part setups.